

Juan Diego Cortés Torres

Backend Developer | .NET / C#

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PROFILE

Backend developer with a degree in Systems and Computer Engineering, specialized in C# (.NET) and REST API development. I have designed and deployed projects using layered and microservices architectures, JWT authentication, relational database integration, and CI/CD pipelines. I focus on writing clean, well-structured, maintainable code that is production-ready from the first commit.

EXPERIENCE

Programming Teaching Assistant — Universidad del Quindío

2024

- Supported students struggling with programming fundamentals, tailoring explanations to each student's individual needs and level.
- Explained data structures, algorithmic thinking, and problem-solving using Python as the reference language.
- Reinforced programming best practices and algorithmic thinking through one-on-one tutoring sessions.

PROJECTS

Counselor (2026) | PWA · React 18 · TypeScript · Firebase · Zustand — Real user, production deployment

- Built a real tool for a school counselor managing three campuses: surfaces who needs attention today and who hasn't been seen in too long. Deployed to production and actively used by the client.
- Layered architecture with repository pattern: the UI never talks directly to Firebase. The same codebase runs against Firestore or localStorage via a build flag, so the public demo stays in sync without maintaining a separate fork.
- Optimized for the user's real device (low-end Android): uncontrolled forms, per-route code splitting, and cacheable chunks. Removed 495 kB of Firebase SDK never executed, cutting the precache from 776 to 294 KiB.
- Firestore security rules enforced by whitelist, versioned and validated against the local emulator. Having an active session is not enough to read data — required because the records involve minors.
- Demo: <https://counselorapp.netlify.app> | Repository: <https://github.com/juandic24/Counselor>

JC Soluciones Energéticas (2026) | Next.js 16 · React 19 · TypeScript · Tailwind v4 · Vercel

- Designed and implemented the website and brand identity for a real solar installation business, deployed to production on Vercel.
- Content architecture allows the non-technical owner to manage the site without touching code.
- Live site: <https://jc-soluciones-energeticas.vercel.app> | Repository: <https://github.com/juandic24/jc-soluciones-energeticas>

AnimalLibrary (2026) | REST API — ASP.NET Core · PostgreSQL · Entity Framework Core

- Separated concerns across three layers (Controllers / Services / Repositories) to ensure each component has a single reason to change.
- Used DTOs to decouple the domain model from the presentation layer, preventing internal data from being exposed to clients.
- Implemented fully asynchronous operations with async/await on all PostgreSQL calls, avoiding thread blocking under concurrent load.
- Repository: <https://github.com/juandic24/animalLibrary>

GameStore (2026) | REST Minimal API — .NET · JWT · SQLite

- Implemented JWT authentication with role-based authorization (admin/user), securing sensitive endpoints without coupling security logic to business logic.
- Applied Data Annotations validation and covered the full resource lifecycle with differentiated public and protected endpoints.
- Repository: <https://github.com/juandic24/gameStore>

Microservices Application (2024) | Distributed Architecture — Python · Java/Spring Boot · Docker

- Designed services with a single responsibility (authentication, monitoring, logging, notifications), each independently deployable and scalable.
- Containerized and orchestrated all services with Docker and docker-compose, reproducing the full environment with a single command.
- Repository: <https://github.com/juandic24/proyecto-final-microservicios>

DevOps Demo Pipeline (2026) | Jenkins · Docker · Bash · LocalStack

- Automated the full build → test → deploy cycle with Jenkins, eliminating manual intervention on every push to the repository.
- Simulated AWS infrastructure (ECS, ECR, VPC) with LocalStack to validate the pipeline without real cloud costs.
- Repository: <https://github.com/juandic24/devops-demo>

Modern Pong (2025) | 2D Game — Unity / C# · itch.io

- Built a complete game with singleplayer (AI) and local multiplayer modes, including gameplay logic, UI system, and custom pixel art and audio assets.
- Play online: <https://juandc25.itch.io/modern-pong> | Repository: <https://github.com/juandic24/modern-pong>

SKILLS

Languages: C#, Java, Python, SQL, JavaScript, TypeScript

Backend: .NET, ASP.NET Core, REST APIs, Entity Framework Core, Spring Boot

Frontend: React, Next.js, Tailwind CSS, HTML, CSS

Firebase: Firestore, Authentication, security rules, local emulator, PWA

Testing: xUnit, Moq (unit and integration testing)

Databases: PostgreSQL, MySQL, SQL Server, SQLite

DevOps: Docker, Jenkins, GitHub Actions, LocalStack

Tools: Git, GitHub, Postman

Languages: Spanish (Native), English (B2 — EF SET 62)

EDUCATION

Universidad del Quindío

B.Sc. in Systems and Computer Engineering

2019 – 2025

Armenia, Colombia